EE5311: Digital IC Design
Tutorial 6 Report

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Roll Number: EE22B032

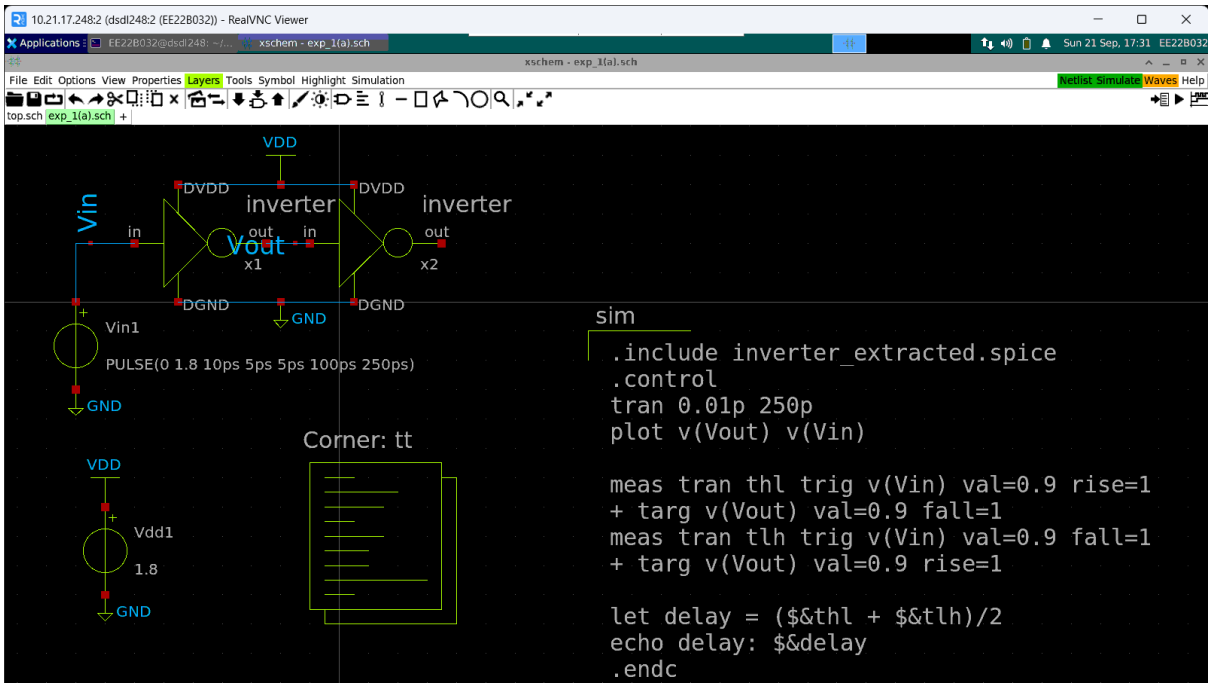
September 21, 2025

Question 1

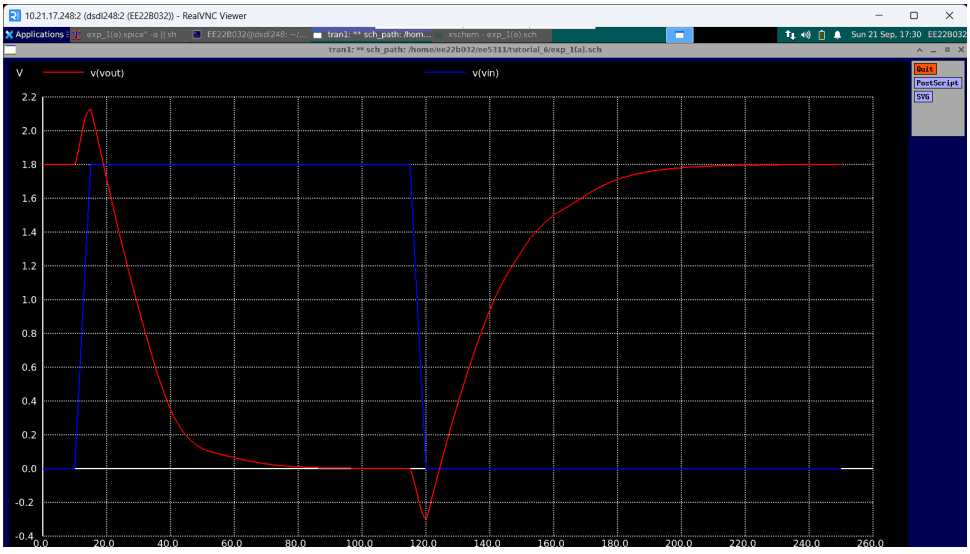
Layout



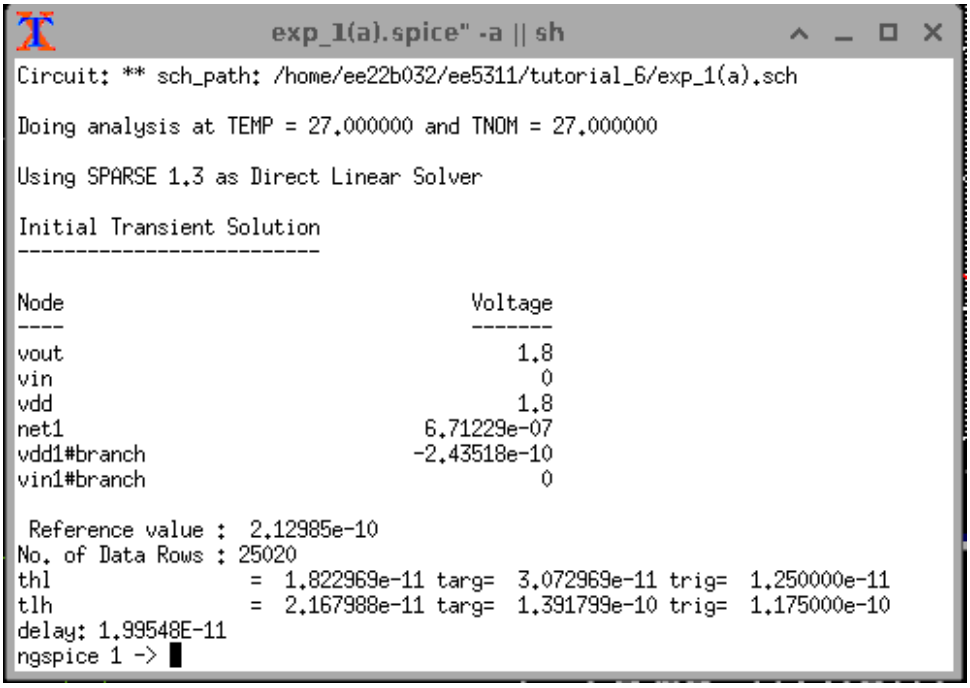
Schematic



Without Parasitics

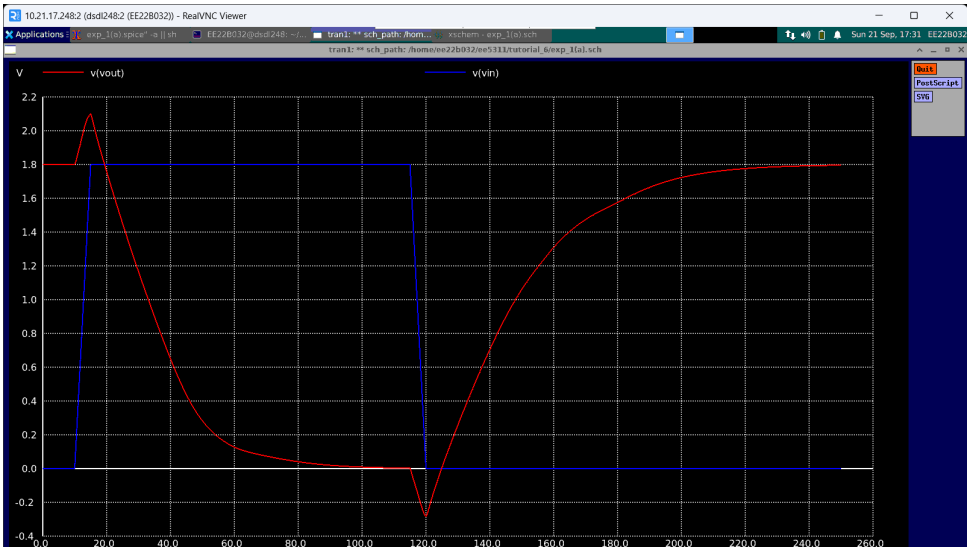


Graph

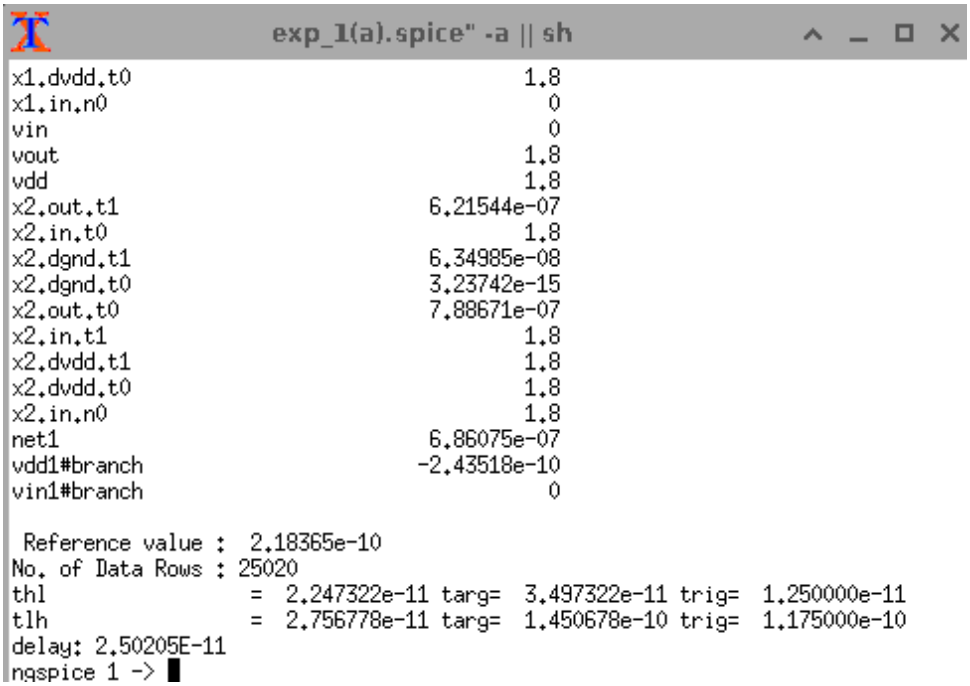


Delay

With Parasitics

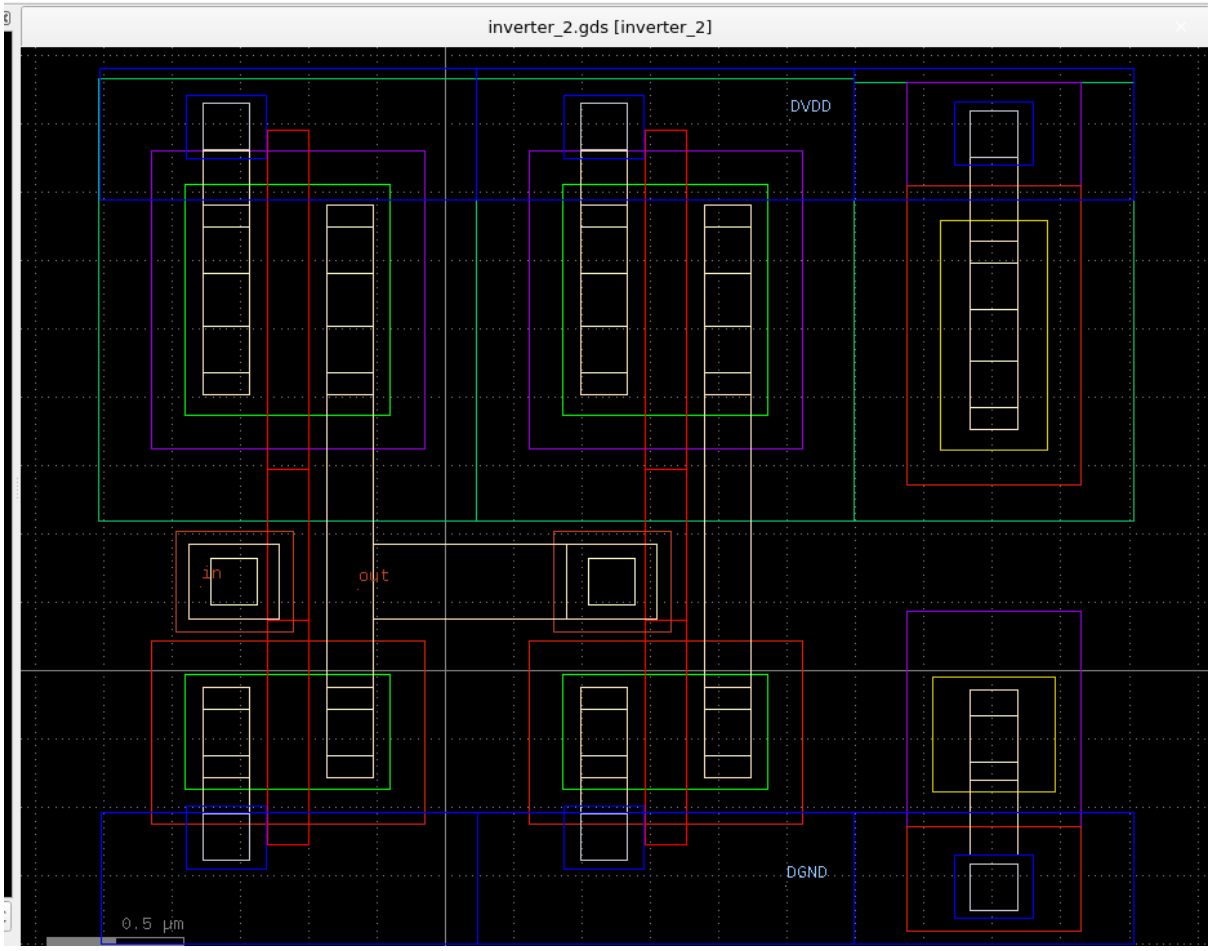


Graph


exp_1(a).spice" -a || sh
x1.dvdd.t0 1.8
x1.in.n0 0
vin 0
vout 1.8
vdd 1.8
x2.out.t1 6.21544e-07
x2.in.t0 1.8
x2.dgnd.t1 6.34985e-08
x2.dgnd.t0 3.23742e-15
x2.out.t0 7.88671e-07
x2.in.t1 1.8
x2.dvdd.t1 1.8
x2.dvdd.t0 1.8
x2.in.n0 1.8
net1 6.86075e-07
vdd1#branch -2.43518e-10
vin1#branch 0
Reference value : 2.18365e-10
No. of Data Rows : 25020
thl = 2.247322e-11 targ= 3.497322e-11 trig= 1.250000e-11
tlh = 2.756778e-11 targ= 1.450678e-10 trig= 1.175000e-10
delay: 2.50205E-11
ngspice 1 ->

Delay

Part B

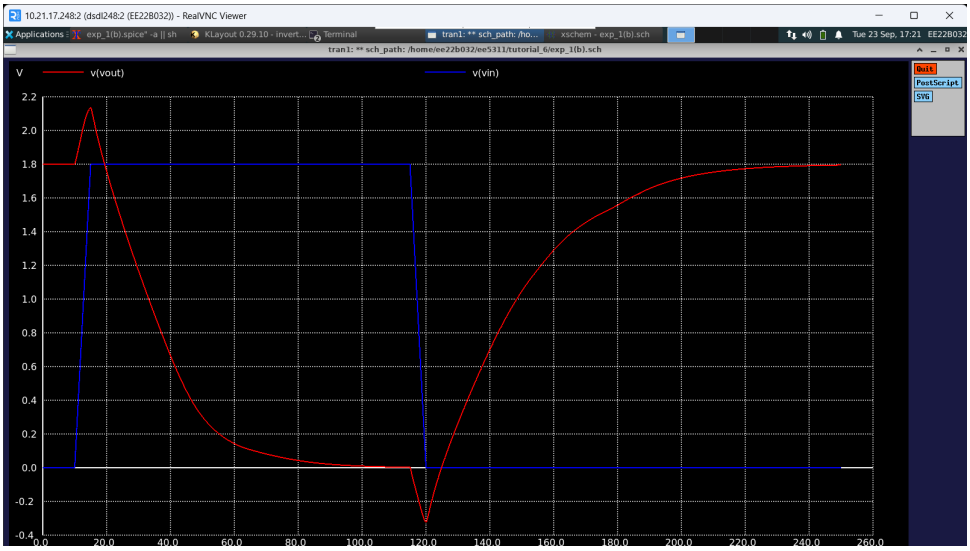


Layout

```
exp_1(b).spice" -a || sh
x1.out.t3                       1.8
x1.dgnd.t1                  6.34985e-08
x1.dgnd.t0                  8.34448e-09
x1.out.t0                       1.8
x1.in.t1                          0
x1.dgnd.t3                  5.12314e-11
x1.dgnd.t2                  1.54214e-08
x1.out.n1                       1.8
x1.out.n0                       1.8
vout                            1.8
vdd                             1.8
x1.dvdd.n0                      1.8
x1.in.n0                          0
vin                               0
x1.dgnd.n0                  4.19807e-11
vdd1#branch                -2.43518e-10
vin1#branch                       0

 Reference value :  2.34575e-10
No. of Data Rows : 25020
thl                =  2.266238e-11 targ=  3.516238e-11 trig=  1.250000e-11
tlh                =  2.784821e-11 targ=  1.453482e-10 trig=  1.175000e-10
delay: 2.52553E-11
ngspice 1 ->
```

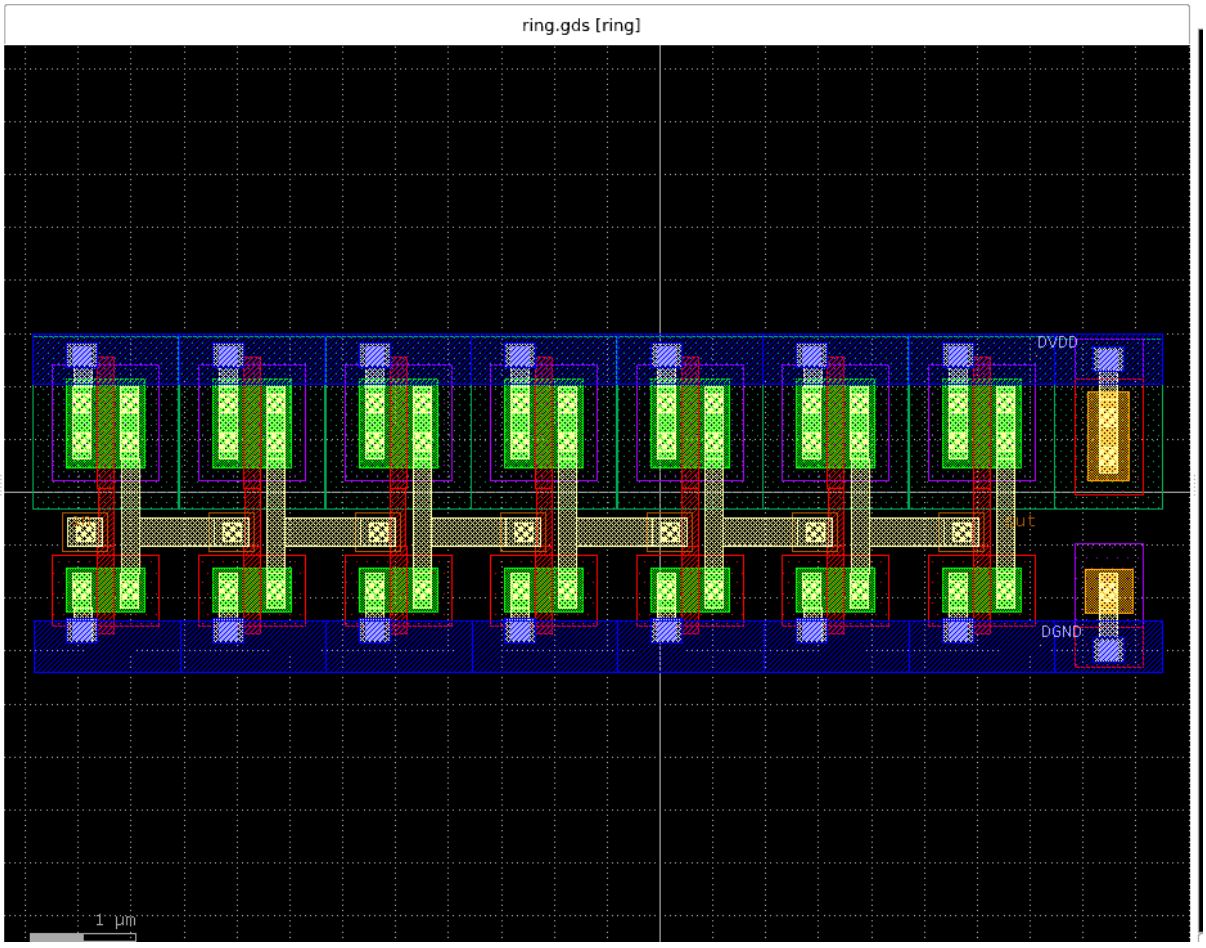
Delay



Graph

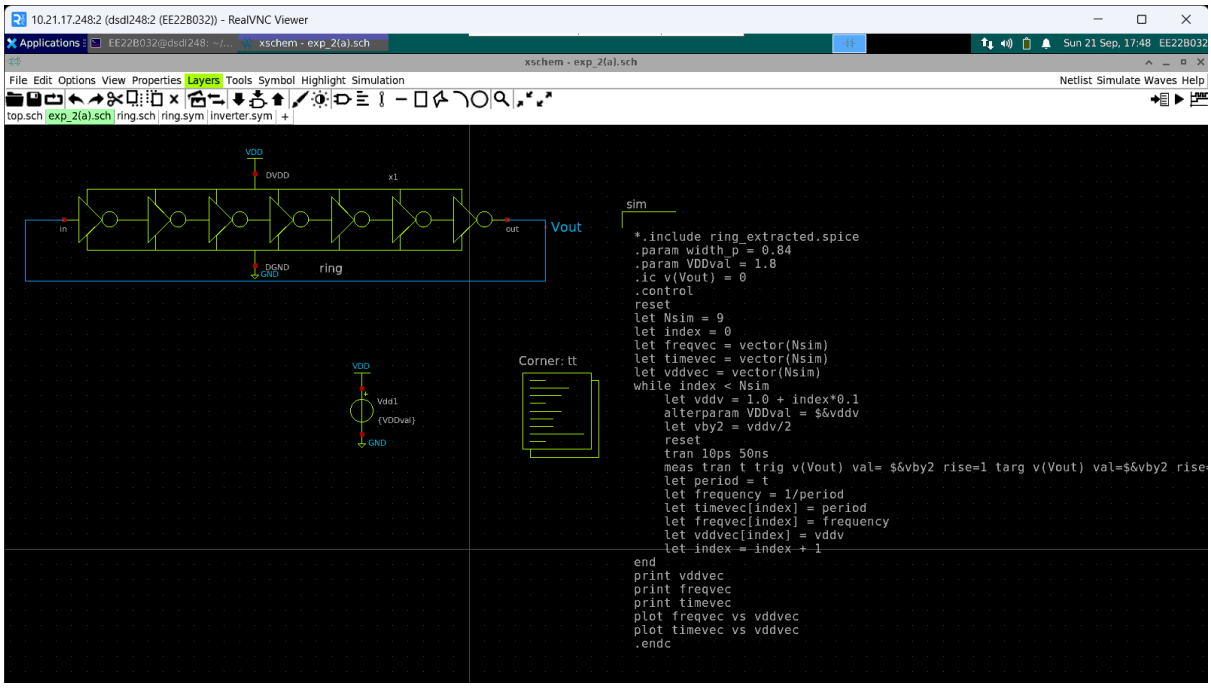
Question 2

Layout



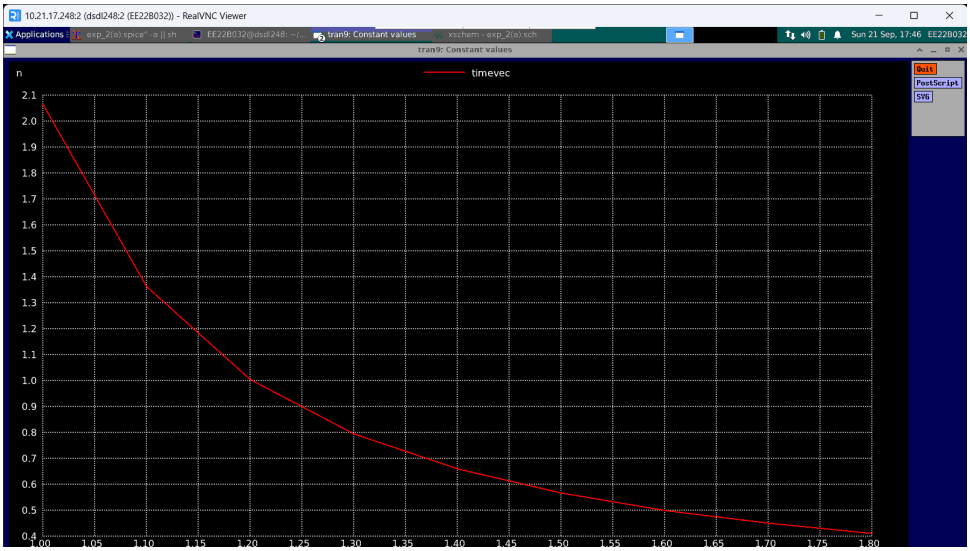
Layout

Schematic

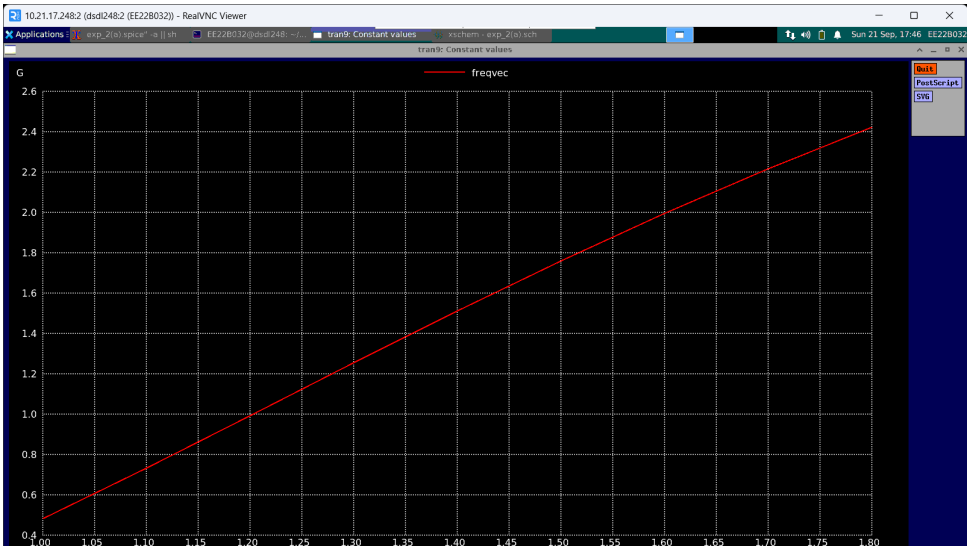


Schematic

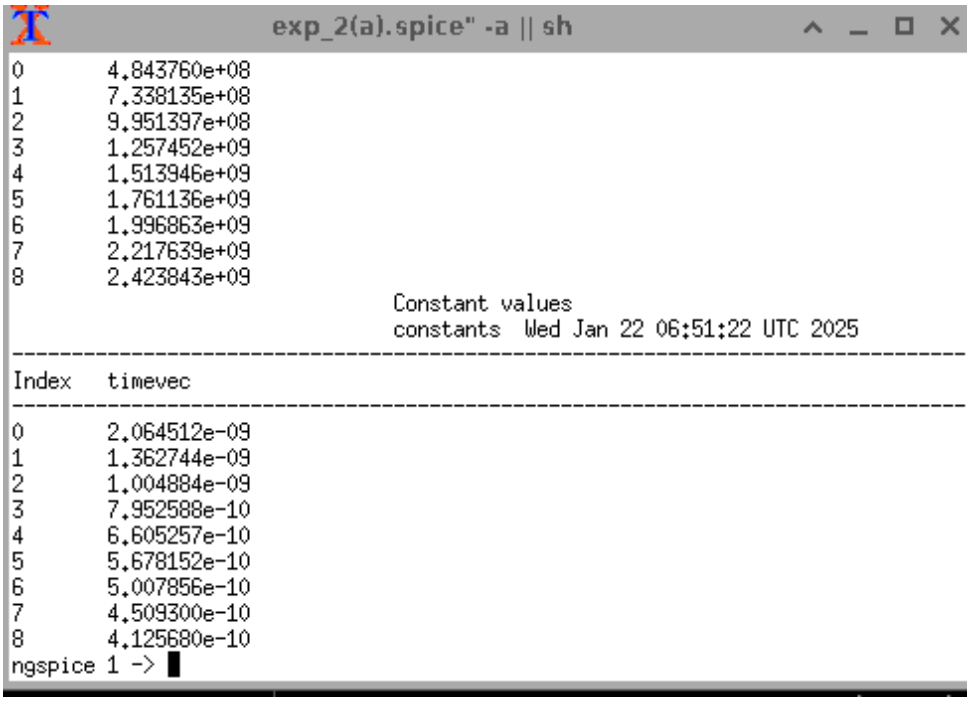
Without Parasitics



Time

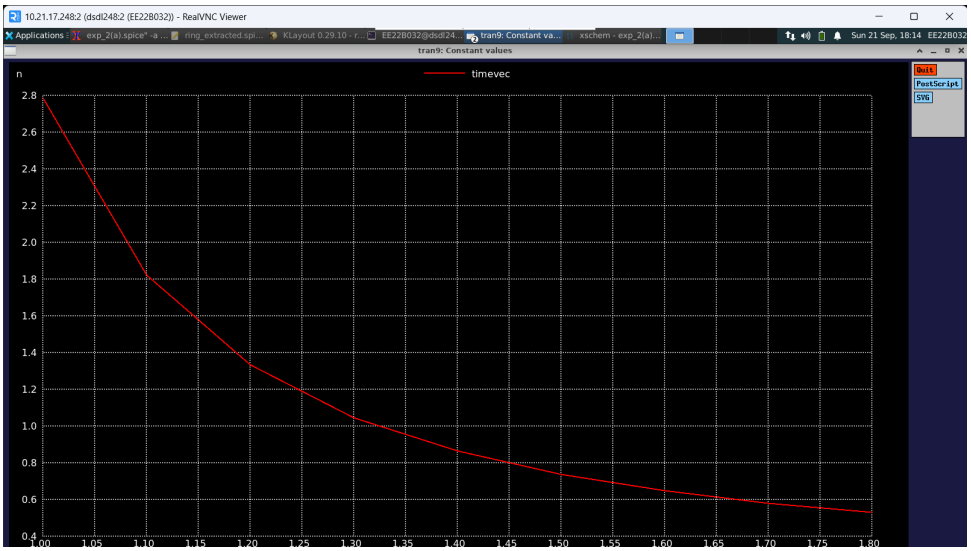


Frequency

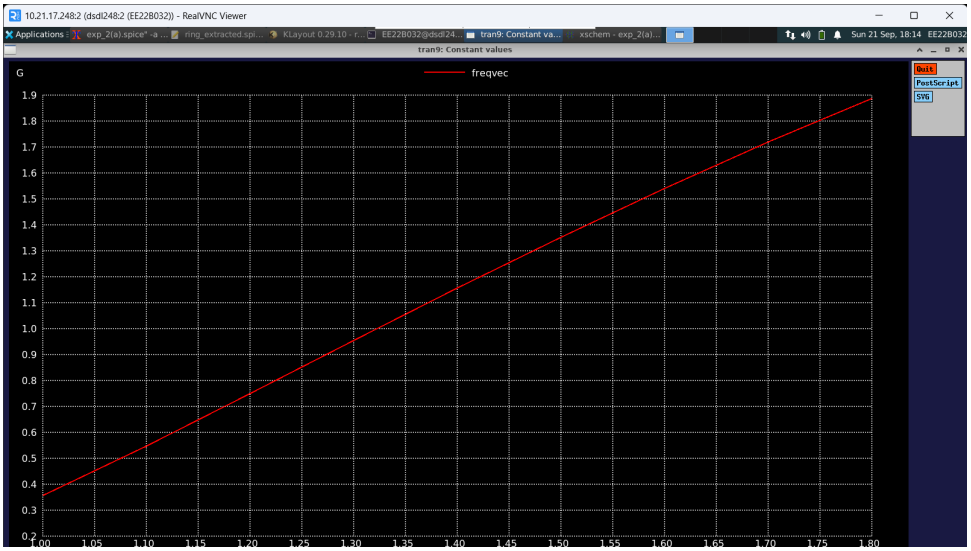


values

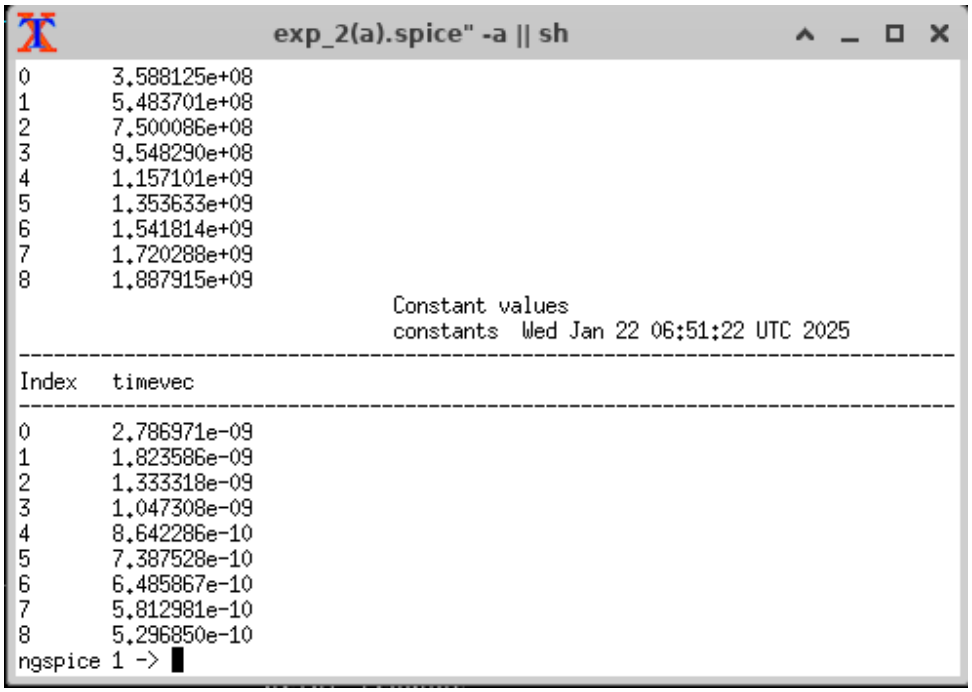
With Parasitics



Time



Frequency


exp_2(a).spice" -a || sh
0 3.588125e+08
1 5.483701e+08
2 7.500086e+08
3 9.548290e+08
4 1.157101e+09
5 1.353633e+09
6 1.541814e+09
7 1.720288e+09
8 1.887915e+09
Constant values
constants Wed Jan 22 06:51:22 UTC 2025
Index timevec
0 2.786971e-09
1 1.823586e-09
2 1.333318e-09
3 1.047308e-09
4 8.642286e-10
5 7.387528e-10
6 6.485867e-10
7 5.812981e-10
8 5.296850e-10
ngspice 1 ->

values